

Smart Queue Management System

1. Problem Statement

Traditional queue management systems in hospitals, banks, colleges, government offices, and other service centers rely on manual token distribution and physical waiting lines. This often leads to long waiting times, overcrowding, lack of transparency, and poor customer experience. Customers have no clear information about their queue status or estimated waiting time, while administrators face difficulties in managing queues efficiently and maintaining service records. Therefore, there is a need for a digital Smart Queue Management System that enables online token booking, real-time queue tracking, and efficient queue management to improve service quality and reduce customer waiting time.

2. Objectives

Primary Objective

To develop a web-based Smart Queue Management System that simplifies queue management, reduces customer waiting time, and improves service efficiency.

Specific Objectives

To provide secure user registration and login.

To enable online token booking for customers.

To display real-time queue status and estimated waiting time.

To provide an administrator dashboard for efficient queue management.

To reduce overcrowding and unnecessary physical waiting.

To maintain queue history and generate service reports.

To improve customer satisfaction through a transparent and organized queue system.

3. Scope

The Smart Queue Management System is designed for organizations that manage a large number of customers every day. It can be implemented in hospitals, banks,

colleges, government offices, clinics, restaurants, salons, and other service centers. The system focuses on online token generation, real-time queue tracking, queue management, and reporting. It aims to improve operational efficiency, reduce manual work, and provide a convenient experience for both customers and administrators. Future enhancements may include mobile applications, QR code-based tokens, and SMS or email notifications.

4. Expected Outcomes

Reduced customer waiting time and improved service efficiency.

Digital queue management with online token booking.

Real-time queue status and estimated waiting time for users.

Reduced overcrowding in waiting areas.

Improved customer satisfaction through transparent queue management.

Efficient administration with accurate queue records and reports.